

SPANGDAHLEM AB, Germany

WMO: 106070

Lat: 49.973N

Lon: 6.693E

Elev: 365

StdP: 97.02

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-8.2	-6.2	-11.9	1.4	-5.8	-9.2	1.8	-4.9	12.1	6.9	10.9	6.9	3.4	70	0.565

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.1	29.8	19.3	27.5	18.4	25.7	17.7	20.4	27.0	19.5	25.6	18.5	23.9	3.8	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.2	13.7	22.0	17.2	12.9	21.1	16.6	12.4	20.5	60.2	26.8	56.9	25.2	53.8	24.0	25.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.6	8.5	7.5	DB	-11.2	33.2	2.9	1.8	-13.2	34.5	-14.9	35.6	-16.5	36.6	-18.6	38.0	(4)
(5)			WB	-11.9	22.3	2.8	1.3	-13.9	23.3	-15.6	24.0	-17.1	24.7	-19.2	25.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	9.6	1.3	2.3	5.5	9.4	13.2	16.2	18.2	17.8	14.0	9.8	5.0	2.1		
	DBStd	7.02	4.02	3.88	3.39	4.01	3.68	3.44	3.47	3.13	3.00	3.55	3.74	3.92		
	HDD10.0	1151	270	216	143	59	12	1	0	0	3	48	153	247		
	HDD18.3	3288	528	449	398	269	164	82	47	48	133	266	399	504		
	CDD10.0	1004	0	0	4	41	110	187	254	241	123	41	4	0		
	CDD18.3	101	0	0	0	1	4	19	42	31	4	0	0	0		
	CDH23.3	995	0	0	0	7	48	188	402	304	44	1	0	0		
	CDH26.7	271	0	0	0	0	7	48	125	85	7	0	0	0		
(14) Wind	WSAvg	3.5	4.1	3.9	4.0	3.4	3.3	3.2	3.2	2.9	3.1	3.3	3.5	3.9		
(15) Precipitation	PrecAvg	767	70	57	50	47	65	70	71	66	62	65	66	79		
	PrecMax	1009	197	149	141	123	128	171	194	144	138	183	133	217		
	PrecMin	565	8	5	7	0	13	20	22	10	17	23	4	10		
	PrecStd	109	38	34	32	26	31	37	36	34	35	34	30	42		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.1	12.8	18.9	24.3	27.2	30.9	33.0	32.3	27.8	22.1	15.1	11.2		
		MCWB	9.7	7.9	11.1	14.1	18.0	20.0	20.1	20.3	17.9	15.6	12.9	10.2		
	2%	DB	9.9	10.1	15.9	21.8	24.3	27.9	30.0	29.0	24.2	18.8	12.9	10.0		
		MCWB	8.8	7.4	9.6	13.1	16.2	18.8	19.4	19.0	17.1	14.9	11.3	9.0		
	5%	DB	8.2	8.8	13.1	19.1	22.2	25.2	27.7	26.8	21.9	16.2	11.5	8.8		
		MCWB	7.5	7.2	8.3	12.1	15.0	17.6	18.5	18.1	16.0	13.6	10.3	8.1		
	10%	DB	6.9	7.5	11.0	16.7	20.1	23.1	25.3	24.5	19.5	14.9	10.1	7.2		
		MCWB	6.2	6.5	7.8	10.9	14.3	16.7	17.6	17.4	15.0	12.9	9.1	6.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.5	9.6	11.8	15.6	19.5	21.7	21.6	21.3	19.5	16.7	13.4	10.7		
		MCDB	10.9	10.8	16.5	21.6	24.8	27.4	29.6	28.1	25.2	19.8	14.2	10.9		
	2%	WB	9.0	8.3	10.2	14.0	17.7	19.9	20.3	20.0	17.8	15.4	11.7	9.2		
		MCDB	9.5	9.2	13.9	20.0	22.2	25.5	27.5	26.8	22.8	18.1	12.8	9.7		
	5%	WB	7.5	7.4	9.4	12.7	16.2	18.6	19.4	19.1	16.6	14.2	10.6	8.1		
		MCDB	8.1	8.3	12.2	17.8	20.4	24.0	25.9	24.5	20.7	16.1	11.6	8.5		
	10%	WB	6.0	6.4	8.3	11.5	14.9	17.4	18.3	18.1	15.5	13.1	9.4	6.8		
		MCDB	6.5	7.3	10.5	15.9	19.0	22.1	23.9	23.0	18.8	14.7	10.3	7.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	4.4	5.6	7.8	9.6	9.7	9.7	10.1	10.2	8.8	7.2	4.7	4.2		
		MCDBR	5.5	7.8	12.2	13.4	13.1	13.5	14.0	14.3	12.4	9.6	6.0	5.1		
	5% WB	MCWBR	4.9	5.8	7.1	6.7	6.5	6.5	5.6	6.2	6.4	6.2	4.6	4.7		
		MCWBR	5.2	6.6	9.8	11.8	11.5	11.8	12.3	12.0	10.8	8.6	5.5	5.0		
(40) Clear-Sky Solar Irradiance	taub		0.304	0.330	0.361	0.392	0.400	0.403	0.401	0.397	0.368	0.352	0.325	0.302		
		taud	2.433	2.382	2.315	2.264	2.271	2.285	2.283	2.323	2.381	2.424	2.434	2.423		
	Ebn at Noon		744	808	840	850	858	856	851	837	824	771	706	688		
		Edh at Noon	62	83	105	123	128	127	125	115	98	79	62	55		
(44) All-Sky Solar Radiation	RadAvg		0.79	1.50	2.67	4.21	4.96	5.50	5.21	4.47	3.34	1.93	0.89	0.61		
	RadStd		0.12	0.28	0.41	0.52	0.49	0.50	0.57	0.45	0.41	0.19	0.11	0.09		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (64 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page